AAMIR TALAL

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Education

CY Tech, Paris *2026 – 2029*
Engineering Cycle (M.Eng), Computer Science – Top 5% of the cohort

- Relevant coursework: Machine Learning, Deep Learning, Optimisation, Probability & Statistics, Linear Algebra, Algorithms & Data Structures, Databases

CY Tech, Paris *2024 – 2026*
Integrated Preparatory Cycle (MPSI-equivalent): advanced mathematics, physics, computer science

- Relevant coursework: Real Analysis, Linear Algebra, Probability, Differential Equations, Algorithms in C & Python, Numerical Analysis

Experience

Safran Aircraft Engines *2026 – 2029*
AI Engineer, Applied Research (Apprenticeship) *Montereau-sur-le-Jard*

- Design and deploy LLM agents that automate and digitalise sales administration (ADV) processes
- Build and evaluate agentic workflows that ground generative AI in existing business tools and data
- Work directly with business teams to identify, scope and prioritise automation use cases

Applied Research Projects

Legible: speaker-attributed grounding for legal LLMs *2026*
Winner, Mistral×Stanford Law School Hackathon

- Designed the System 1→System 2 engine: rule-based zoning and a calibrated classifier (Jev) attribute each sentence of a French court ruling to its speaker with a confidence score, conditioning a Mistral LLM that answers with paragraph-level citations
- On the team's 25-ruling benchmark, 4× fewer misattributed answers than a baseline LLM (11 vs 44 out of 224)
- Stack: Python, FastAPI, Mistral API, MCP; live demo: legible-or4n.onrender.com

Pulse-Based Deepfake Detection via rPPG *2026*
Independent reimplementation from the literature (Frontiers, 2025)

- Reimplemented from scratch, from a published paper, a remote photoplethysmography (rPPG) pipeline: MediaPipe face tracking, forehead and cheek ROIs, band-pass filtered pulse estimation (0.7–3 Hz, FFT)
- Validated heart-rate estimation on UBFC, comparing green-channel, CHROM and POS methods (MAE: [X] / [Y] / [Z] BPM)
- Designed window-level pulse-consistency features (cross-region coherence, spectral peak regularity, SNR) and evaluated a classifier with subject-disjoint GroupKFold (AUC [Y])
- Analysed limits: compression, motion, skin-tone bias, and fakes that reproduce a plausible pulse
- Stack: Python, OpenCV, MediaPipe, SciPy, scikit-learn

MiroPolis: multi-agent LLM simulation of parliament *2026*
Top Project, National Assembly Hackathon

- Built a multi-agent digital twin (OASIS) in which LLM deputies, seeded with the chamber's real composition, groups and programmes, debate, amend and vote on a bill
- Compared fine-tuned agents with persona-orchestrated pre-trained LLMs, and pivoted to the latter as faster and more realistic

Technical Skills

Machine Learning	Signal processing (rPPG, FFT), classification, cross-validation, PyTorch
AI / LLM	LLM agents, RAG, prompt engineering, LLM evaluation, MCP, LangChain, Hugging Face
Data	SQL, pandas, NumPy, scikit-learn, data pipelines, n8n
Languages	Python, C, SQL, JavaScript, Shell/Bash, PHP
Spoken	French (native), English (C1), Urdu (C1)

Hackathons & Achievements

- Winner, Mistral×Stanford Law School Hackathon, and winner of the Legora track
- Top Score, IBM×Airbus Hackathon (Télécom Paris)
- Top Project, French National Assembly Hackathon
- Top 20 out of 300, Anthropic×Hugging Face Hackathon
- ML algorithms and maths implemented from scratch on Deep-ML; 50+ LeetCode problems solved

Community & Interests

- Content creator, Intellectuel Athlète (personal development / tech): 14,000+ followers, 2–3M monthly views
- Volunteering: 50+ weekly humanitarian outreach rounds (maraudes), Au cœur de la précarité; created an Arabic-learning web app (1,000+ users in 24 h)
- Sport: Brazilian jiu-jitsu and grappling (competition), running